

MAILFLOW

Product Requirements Document (PRD)

Version 1.0

Status: Draft MVP

Date: 27 July 2026

MailFlow

Product Requirements Document — Lead Outreach & Follow-up Platform (Email + WhatsApp)

Document Owner: Founder / Solo Builder

Status: Draft v1.0 — MVP Definition

Date: 27 July 2026

Build Approach: Solo development using Antigravity (AI-assisted IDE)

1. Overview

MailFlow is a lead outreach and lightweight CRM tool that lets a user upload a list of leads (Excel/CSV) with any set of columns, map those columns to a standard schema, write a single message template with dynamic placeholders, and then reach every lead through two channels: bulk personalized email and one-click, pre-filled WhatsApp messages sent manually per lead. It also tracks contact status and follow-ups so outreach doesn't fall through the cracks.

2. Problem Statement

Small businesses, freelancers, and agencies doing cold outreach (e.g. pitching web design, marketing, or other services to local businesses such as clinics) manage leads in scattered spreadsheets and send messages one-by-one, copy-pasting the same text repeatedly and manually editing names or details. There is no single place to: (a) hold the lead list, (b) personalize a message per lead, (c) send it via email in bulk or via WhatsApp with one click, and (d) track who has been contacted and who needs a follow-up.

3. Goals

Let a user go from a raw Excel/CSV lead list to a sent, personalized campaign in minutes.

Support leads with arbitrary, non-fixed column structures (different industries have different fields).

Make bulk email sending reliable and personalized using a template + placeholder system.

Make WhatsApp outreach fast but manual/safe — one click opens a pre-filled message per lead, no automated mass-send.

Give the user basic CRM visibility: contacted status, follow-up date, and outcome per lead.

Be buildable solo, incrementally, while learning modern backend concepts (queues, workers) along the way.

4. Target Users

Primary (v1): the founder herself, for outbound pitching to local businesses (e.g. clinics in Udaipur).

Secondary (future): freelancers, small agencies, and solo consultants who run their own cold outreach and want a lightweight alternative to expensive cold-email SaaS tools.

5. Scope — MVP (Phase 1)

5.1 In Scope

Upload leads via Excel (.xlsx) or CSV file.

Automatic column detection with a confirmation/mapping screen (auto-guess + manual override).

Core fields: Name, Phone, Email, Status, Contacted (yes/no), Follow-up Date.

Any additional uploaded columns are stored as custom fields per lead (flexible schema, not hardcoded).

Message template builder with placeholder variables drawn from mapped + custom fields (e.g. {{Name}}, {{Area}}).

Bulk email sending using a background job queue (BullMQ + Redis) with per-email status (sent/failed/pending).

WhatsApp outreach: dashboard shows each lead with a WhatsApp icon; clicking opens a pre-filled wa.me link with the personalized message. Sending itself is manual and one-by-one — no automated "send all" for WhatsApp.

Lead status tracking: Not Contacted / Contacted / Follow-up Due / Converted / Rejected.

Simple dashboard: leads table, filters by status, campaign send history.

5.2 Out of Scope (for MVP)

Automated bulk WhatsApp sending (via unofficial APIs or automation) — explicitly excluded due to ban risk and platform policy.

Company data enrichment from third-party sources (e.g. auto-fetching company size, industry) — user supplies this manually in the sheet for now.

PDF lead upload — Excel/CSV only in v1; PDF parsing deferred.

Multi-user / team accounts — single-user usage for MVP.

WhatsApp Business API integration — considered for a later phase if send-volume needs grow.

6. Core Features — Detail

6.1 Lead Upload & Column Mapping

User uploads an .xlsx or .csv file. The system reads the header row and auto-suggests a mapping to core fields using keyword matching (e.g. a header containing "phone", "mobile", or "contact no" maps to Phone). The user reviews and confirms or corrects this mapping on a dedicated screen before leads are imported. Any column that doesn't match a core field is preserved as a custom field on each lead record, so sheets with different structures (clinics, real estate, retail, etc.) are all supported without code changes.

6.2 Template Builder

User writes one message template (used as a base for both email and WhatsApp, editable separately if needed). The editor shows available placeholders based on the current lead sheet's mapped and custom fields (e.g. {{Name}}, {{Doctor}}, {{Area}}), inserted into the message text and substituted per-lead at send time.

6.3 Bulk Email Sending

On send, one background job per lead is added to a BullMQ queue backed by Redis. A separate worker process picks up jobs, calls the email provider (Resend or SendGrid) with the personalized content, applies basic rate limiting, and updates each lead's send status (sent / failed / pending) in the database, with automatic retry on transient failures.

6.4 WhatsApp Outreach (Manual, One-by-One)

For each lead, the dashboard shows a WhatsApp icon. Clicking it generates a wa.me link with the phone number and the personalized message pre-filled, opening WhatsApp Web or the WhatsApp app for the user to review and send manually. No message is sent without the user's own action, and there is no bulk/automated send button for this channel.

6.5 CRM / Status Tracking

Each lead has a status field (Not Contacted, Contacted, Follow-up Due, Converted, Rejected) and a follow-up date. The dashboard supports filtering by status so the user can quickly see who to follow up with next.

7. User Flow (High Level)

- 1. User uploads Excel/CSV file of leads.**
- 2. System auto-detects columns and shows a mapping confirmation screen.**
- 3. User confirms/edits mapping; leads are imported into the database.**
- 4. User writes a message template, inserting placeholders from available fields.**

5a. Email path: user clicks "Send Campaign" → jobs queued → worker sends emails in the background → dashboard shows live status per lead.

5b. WhatsApp path: user opens the dashboard, clicks the WhatsApp icon per lead → pre-filled chat opens → user reviews and sends manually.

6. User updates lead status and follow-up date as replies come in or calls are made.

8. Data Model (Conceptual)

9. Tech Stack

10. Non-Functional Requirements

Reliability: email sends should retry automatically on transient failure; failed sends must be visible, not silent.

Data safety: uploaded lead data (contact details) should be stored securely; access limited to the account owner.

Usability: column mapping and template building should require no technical knowledge from the end user.

Compliance-awareness: outreach should support easy identification of consent/opt-out where applicable, and avoid patterns that resemble spam or automated messaging abuse.

11. Risks & Considerations

Email deliverability: bulk sending from a new domain risks landing in spam without proper SPF/DKIM/DMARC setup and sender warm-up.

WhatsApp platform risk: any automated or scripted bulk sending (even disguised as "one click, many messages") risks number bans under WhatsApp's terms; the manual, one-by-one design in this PRD is intended to avoid that risk.

Legal/regulatory: unsolicited commercial email and messaging may be subject to regulations (e.g. India's IT Act, and CAN-SPAM/GDPR if contacting leads outside India); the product should make it easy to honor opt-outs.

Data quality: since company/lead detail enrichment is manual (not automated) in MVP, personalization quality depends entirely on what the user has filled into the sheet.

12. Success Metrics (MVP)

Time to go from "upload file" to "first message sent" (target: under 10 minutes for a new user).

% of uploaded columns correctly auto-mapped without manual correction.

Email delivery success rate (sent vs. failed) per campaign.

Number of leads moved from "Not Contacted" to "Contacted"/"Follow-up Due" per week, as a proxy for actual usage.

13. Future Roadmap (Post-MVP)

Phase 2: PDF lead upload support.

Phase 2: Optional company/domain enrichment (e.g. fetching a logo or basic public info) via a paid enrichment API.

Phase 3: WhatsApp Business API integration for approved-template bulk sending, if volume justifies the cost and setup.

Phase 3: Multi-user/team accounts, shared lead pools, and role-based access.

Phase 3: Analytics dashboard (open rates, reply rates, conversion funnel).

14. Open Questions

Should Email be a required field per lead, or should WhatsApp-only leads (no email) be fully supported as a first-class case?

What is the expected lead volume per campaign (tens, hundreds, thousands)? This affects how urgently rate-limiting and email domain warm-up need to be addressed.

Will the tool remain single-user, or is there near-term interest in eventually offering it to other freelancers/agencies as a product?